AP Precalculus **Unit 4, Lesson 4.2**

Homework — Answer Key

Name: ____________________________ Date: ________________ Period: _________

Parametric Planar Motion Practice — Key

1. Let $f(t) = (t^2 - 2t,\ 3t - t^2)$ for $0 \le t \le 4$.

(a) Table:

t	$x(t) = t^2 - 2t$	$y(t) = 3t - t^2$	$(x,\ y)$
0	0	0	$(0,\ 0)$
1	-1	2	$(-1,\ 2)$
2	0	2	$(0,\ 2)$
3	3	0	$(3,\ 0)$
4	8	-4	$(8,\ -4)$

(b) Horizontal extrema:
$x(t) = t^2 - 2t$; vertex at $t = 1$. $x(1) = -1$ (minimum, leftmost position).
Endpoints: $x(0) = 0$, $x(4) = 8$ (rightmost). Maximum is $x = 8$.

(c) Vertical extrema:
$y(t) = 3t - t^2 = -(t^2 - 3t)$; vertex at $t = 3/2$.
$y(3/2) = 9/2 - 9/4 = 9/4 = 2.25$ (maximum, highest position).
Endpoints: $y(0) = 0$, $y(4) = -4$ (minimum, lowest position).

2.

(a) x-intercepts (zeros of $y(t)$):
$3t - t^2 = t(3 - t) = 0 \ \Rightarrow\ t = 0$ or $t = 3$.
$x(0) = 0$, $x(3) = 3$. x-intercepts: $(0,\ 0)$ and $(3,\ 0)$.

(b) y-intercepts (zeros of $x(t)$):
$t^2 - 2t = t(t - 2) = 0 \ \Rightarrow\ t = 0$ or $t = 2$.
$y(0) = 0$, $y(2) = 2$. y-intercepts: $(0,\ 0)$ and $(0,\ 2)$.

Contextual Problem — Key

3. $x(t) = 4t$, $y(t) = -5t^2 + 20t$, $0 \le t \le 4$.

(a) Maximum height:
$y(t) = -5t^2 + 20t$; vertex at $t = -20/(2 \cdot (-5)) = 2$ s.
$y(2) = -20 + 40 = 20$ m. The ball reaches a maximum height of 20 meters at $t = 2$ seconds.

(b) When does the ball hit the ground?
$-5t^2 + 20t = 0 \ \Rightarrow\ -5t(t - 4) = 0 \ \Rightarrow\ t = 4$ s (excluding $t = 0$).
$x(4) = 4 \cdot 4 = 16$ m east. The ball lands at $x = 16$ m east.

AP-Style Practice — Key

4. $h(t) = (t^2 - 2t - 3,\ t^2 - 4)$, $-1 \le t \le 4$. Which correctly identifies the y-intercepts?

(A) $(-4,\ 0)$ and $(5,\ 0)$

(B) **$(0,\ -3)$ and $(0,\ 5)$ $\leftarrow$ correct**

(C) $(0,\ -3)$ only

(D) $(0,\ -4)$ and $(0,\ 12)$

Teacher Note

Solution: y-intercepts occur where $x(t) = 0$.
$t^2 - 2t - 3 = (t-3)(t+1) = 0 \Rightarrow t = 3$ or $t = -1$. Both are in $[-1, 4]$.
At $t = -1$: $y(-1) = 1 - 4 = -3 \to (0, -3)$.
At $t = 3$: $y(3) = 9 - 4 = 5 \to (0, 5)$.
Answer: **(B)**.
Distractor analysis:
(A) $(-4, 0)$ and $(5, 0)$: student found x-intercepts (zeros of $y(t)$) instead of y-intercepts — the classic rule-reversal error.
(C) $(0, -3)$ only: student excluded $t = -1$ because they treated the domain as open or forgot to check all zeros.
(D) Evaluated y at wrong t-values or made arithmetic errors.
Post the rule on the board: *zeros of $x(t)$ $\rightarrow$ y-intercepts of the curve; zeros of $y(t)$ $\rightarrow$ x-intercepts.* If most students choose (A), spend two minutes restating this rule before moving on.